

FOODSY

Database Design Document

V 1.0

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Contents

1	Project Overview	2
1.1	Introduction	2
1.2	Problem Statement	2
1.3	Project Objectives	2
2	Detailed Database Design	3
2.1	Entity	3
2.2	Data Dictionary	4
2.3	Business Rules	9
2.4	Entity Relationship Diagram	10
	Appendix: AI Usage and Prompts	12

CHAPTER 1: PROJECT OVERVIEW

1.1. INTRODUCTION

Foodsy is an online food ordering and delivery management system developed for the restaurant and e-commerce domain. The purpose of this system is to help customers order food easily from restaurants and receive deliveries at their location. The system supports important activities such as menu browsing, cart management, order placement, payment handling (via third party), delivery assignment, and live order tracking. It is designed for customers, restaurant admins, and delivery riders to work together on one platform. Foodsy also includes features like promo codes, notifications, reviews, and delivery area management. The system can be used by restaurants and food delivery businesses to improve customer service and manage operations efficiently. The main goal of the system is to provide a fast, organized, and user-friendly food delivery experience.

1.2. PROBLEM STATEMENT

In the food delivery domain, many restaurants still face problems in managing customer orders, deliveries, payments, and menu records efficiently. Manual handling of orders can cause delays, missing information, and incorrect order processing. Customers may experience difficulties in tracking their orders and receiving timely updates. Restaurants also face challenges in managing delivery riders, customer reviews, and promotional offers in an organized way. Existing systems may not provide proper integration between ordering, billing, and delivery processes, which increases data redundancy and reduces efficiency. Without a centralized system, maintaining customer records, payment details, and delivery tracking becomes difficult and time-consuming. These issues can negatively affect customer satisfaction, business performance, and overall service quality. Foodsy is proposed to solve these problems by providing a complete and organized food ordering and delivery management system.

1.3. PROJECT OBJECTIVES

General Objective:

The main objective of Foodsy is to develop an efficient and user-friendly food ordering and delivery management system that helps restaurants manage orders, customers, payments, and deliveries in an organized and automated way.

Specific Objectives:

- To allow customers to browse food categories and menu items online easily.
- To provide a cart system for managing selected food items before placing orders.
- To automate the order placement and billing process for faster service.
- To manage customer, admin, and rider information in a centralized database.
- To support online payment handling and bill generation.
- To assign delivery riders and track delivery status efficiently.
- To provide live delivery tracking for improving customer experience.
- To manage promo codes, discounts, and customer notifications effectively.
- To collect customer reviews and ratings for service improvement.
- To reduce manual work, data redundancy, and order processing errors in restaurant operations.

GitHub Repository: <https://github.com/bilal-512/Foodsy>

CHAPTER 2: DETAILED DATABASE DESIGN

2.1. ENTITY

Sr.	Entity Name	Description
01	Person	A person is an individual whose basic information is stored in the Foodsy system, including customers, admins, and riders.
02	Customer	A customer is a user who browses menus, places food orders, makes payments, and gives reviews in the system.
03	Admin	An admin is responsible for managing orders, riders, notifications, and overall restaurant operations in the system.
04	Rider	A rider is responsible for delivering food orders from the restaurant to customers.
05	Restaurant	A restaurant represents the food business that provides menu items and delivery services through Foodsy.
06	Category	A category is used to organize menu items into groups such as burgers, pizzas, or drinks.
07	Menu Item	A menu item represents a food product that customers can order from the restaurant.
08	Menu Item Variant	A menu item variant represents different sizes or options available for a menu item.
09	Cart	A cart stores the food items selected by a customer before placing an order.
10	Address	An address stores the delivery location information of customers.
11	Area	An area represents a delivery region used for managing deliveries and rider assignments.
12	Order	An order represents a food purchase made by a customer in the system.
13	Bill	A bill stores the payment details and total charges generated for an order.
14	Payment	A payment represents the transaction made by a customer for an order bill.
15	Promo Code	A promo code provides discounts and special offers to customers during ordering.
16	Delivery	A delivery represents the process of sending food orders from the restaurant to customers.
17	Delivery Tracking	Delivery tracking stores the live location updates of riders during deliveries.
18	Review	A review stores customer feedback and ratings about food and delivery services.

Sr.	Entity Name	Description
19	Notification	A notification stores messages and updates sent to customers, admins, or riders.

2.2. DATA DICTIONARY

2.2.1 Person

Sr.	Name	Data Type	Constraint	Description
01	PersonID	Integer	Primary Key, NOT NULL	Unique ID of person
02	Name	String	NOT NULL	Name of person
03	Email	String	UNIQUE, NOT NULL	Email address
04	Phone	String	UNIQUE, NOT NULL	Contact number
05	Password	String	NOT NULL	Account password
06	Role	String	NOT NULL	Role of person
07	isActive	Boolean	DEFAULT TRUE	Account status

2.2.2 Admin

Sr.	Name	Data Type	Constraint	Description
01	AdminID	Integer	Primary Key, NOT NULL	Unique admin ID
02	Permissions	String	NOT NULL	Access permissions
03	LastLogin	DateTime	NULL Allowed	Last login time

2.2.3 Customer

Sr.	Name	Data Type	Constraint	Description
01	CustomerID	Integer	Primary Key, NOT NULL	Unique customer ID
02	WalletBalance	Decimal	DEFAULT 0	Customer wallet amount

2.2.4 Rider

Sr.	Name	Data Type	Constraint	Description
01	RiderID	Integer	Primary Key, NOT NULL	Unique rider ID

Sr.	Name	Data Type	Constraint	Description
02	VehicleType	String	NOT NULL	Rider vehicle type
03	Licence	String	UNIQUE	Licence number
04	isAvailable	Boolean	DEFAULT TRUE	Rider availability
05	CurrentLatitude	Decimal	NULL Allowed	Current rider latitude
06	CurrentLongitude	Decimal	NULL Allowed	Current rider longitude
07	AvgRating	Decimal	Range 1–5	Average rider rating

2.2.5 Restaurant

Sr.	Name	Data Type	Constraint	Description
01	RestaurantID	Integer	Primary Key, NOT NULL	Unique restaurant ID
02	Name	String	NOT NULL	Restaurant name
03	Logo	String	NULL Allowed	Restaurant logo
04	OpenTime	Time	NOT NULL	Opening time
05	CloseTime	Time	NOT NULL	Closing time
06	BaseLocation	String	NOT NULL	Restaurant location

2.2.6 Category

Sr.	Name	Data Type	Constraint	Description
01	CategoryID	Integer	Primary Key, NOT NULL	Unique category ID
02	Name	String	NOT NULL	Category name
03	Description	String	NULL Allowed	Category details
04	DisplayOrder	Integer	DEFAULT 0	Category display order
05	isActive	Boolean	DEFAULT TRUE	Category status
06	ImageURL	String	NULL Allowed	Category image

2.2.7 MenuItem

Sr.	Name	Data Type	Constraint	Description
01	MenuItemID	Integer	Primary Key, NOT NULL	Unique menu item ID
02	Name	String	NOT NULL	Food item name
03	Description	String	NULL Allowed	Item details
04	isAvailable	Boolean	DEFAULT TRUE	Item availability
05	BasePrice	Decimal	NOT NULL	Base item price

2.2.8 MenuItemVariant

Sr.	Name	Data Type	Constraint	Description
01	VariantID	Integer	Primary Key, NOT NULL	Unique variant ID
02	Label	String	NOT NULL	Variant name
03	PriceDelta	Decimal	DEFAULT 0	Extra variant price

2.2.9 Cart

Sr.	Name	Data Type	Constraint	Description
01	CartID	Integer	Primary Key, NOT NULL	Unique cart ID
02	Status	String	NOT NULL	Cart status
03	CreatedAt	DateTime	NOT NULL	Cart creation date
04	UpdatedAt	DateTime	NOT NULL	Last updated time

2.2.10 CartItem

Sr.	Name	Data Type	Constraint	Description
01	CartItemID	Integer	Primary Key, NOT NULL	Unique cart item ID
02	Quantity	Integer	NOT NULL	Selected quantity
03	Price	Decimal	NOT NULL	Item total price

2.2.11 Address

Sr.	Name	Data Type	Constraint	Description
01	AddressID	Integer	Primary Key, NOT NULL	Unique address ID
02	Label	String	NOT NULL	Address label
03	Street	String	NOT NULL	Street name
04	City	String	NOT NULL	City name
05	Latitude	Decimal	NULL Allowed	Location latitude
06	Longitude	Decimal	NULL Allowed	Location longitude

2.2.12 Area

Sr.	Name	Data Type	Constraint	Description
01	AreaID	Integer	Primary Key, NOT NULL	Unique area ID

Sr.	Name	Data Type	Constraint	Description
02	AreaName	String	NOT NULL	Delivery area name
03	LatCoordinate	Decimal	NULL Allowed	Area latitude
04	LongCoordinate	Decimal	NULL Allowed	Area longitude

2.2.13 PromoCode

Sr.	Name	Data Type	Constraint	Description
01	PromoCodeID	Integer	Primary Key, NOT NULL	Unique promo ID
02	Code	String	UNIQUE, NOT NULL	Promo code
03	DiscountType	String	NOT NULL	Discount category
04	DiscountValue	Decimal	NOT NULL	Discount amount
05	MinOrderAmount	Decimal	DEFAULT 0	Minimum order value
06	MaxUses	Integer	NOT NULL	Maximum usage limit
07	UsedCount	Integer	DEFAULT 0	Used promo count
08	ValidFrom	DateTime	NOT NULL	Promo start date
09	ValidUntil	DateTime	NOT NULL	Promo expiry date
10	isActive	Boolean	DEFAULT TRUE	Promo status

2.2.14 Orders

Sr.	Name	Data Type	Constraint	Description
01	OrderID	Integer	Primary Key, NOT NULL	Unique order ID
02	OrderStatus	String	NOT NULL	Current order status
03	PlacedAt	DateTime	NOT NULL	Order placement time
04	ConfirmedAt	DateTime	NULL Allowed	Confirmation time
05	ReadyAt	DateTime	NULL Allowed	Order ready time
06	DeliveredAt	DateTime	NULL Allowed	Delivery completion time
07	CancelledAt	DateTime	NULL Allowed	Cancellation time
08	OrderCancelReasonString		NULL Allowed	Cancellation reason

2.2.15 OrderItem

Sr.	Name	Data Type	Constraint	Description
01	OrderItemID	Integer	Primary Key, NOT NULL	Unique order item ID
02	Quantity	Integer	NOT NULL	Ordered quantity
03	UnitPrice	Decimal	NOT NULL	Price per item

2.2.16 Bill

Sr.	Name	Data Type	Constraint	Description
01	BillID	Integer	Primary Key, NOT NULL	Unique bill ID
02	Subtotal	Decimal	NOT NULL	Order subtotal
03	TaxAmount	Decimal	DEFAULT 0	Applied tax
04	DiscountAmount	Decimal	DEFAULT 0	Discount amount
05	DeliveryFee	Decimal	DEFAULT 0	Delivery charges
06	TotalAmount	Decimal	NOT NULL	Final payable amount
07	PaymentStatus	String	NOT NULL	Payment condition
08	GeneratedAt	DateTime	NOT NULL	Bill generation time

2.2.17 Payment

Sr.	Name	Data Type	Constraint	Description
01	PaymentID	Integer	Primary Key, NOT NULL	Unique payment ID
02	Method	String	NOT NULL	Payment method
03	TransactionRef	String	UNIQUE	Transaction reference
04	GatewayResponse	String	NULL Allowed	Payment response
05	PaidAt	DateTime	NOT NULL	Payment date

2.2.18 Delivery

Sr.	Name	Data Type	Constraint	Description
01	DeliveryID	Integer	Primary Key, NOT NULL	Unique delivery ID
02	PickupLat	Decimal	NULL Allowed	Pickup latitude
03	PickupLong	Decimal	NULL Allowed	Pickup longitude
04	DropLat	Decimal	NULL Allowed	Drop latitude
05	DropLong	Decimal	NULL Allowed	Drop longitude
06	DistanceKM	Decimal	NOT NULL	Delivery distance
07	AssignedAt	DateTime	NOT NULL	Assignment time
08	PickupTime	DateTime	NULL Allowed	Pickup time

Sr.	Name	Data Type	Constraint	Description
09	DeliveredTime	DateTime	NULL Allowed	Delivery completion time
10	Status	String	NOT NULL	Delivery status

2.2.19 DeliveryTracking

Sr.	Name	Data Type	Constraint	Description
01	TrackingID	Integer	Primary Key, NOT NULL	Unique tracking ID
02	Latitude	Decimal	NOT NULL	Rider latitude
03	Longitude	Decimal	NOT NULL	Rider longitude
04	RecordedAt	DateTime	NOT NULL	Tracking record time

2.2.20 Review

Sr.	Name	Data Type	Constraint	Description
01	ReviewID	Integer	Primary Key, NOT NULL	Unique review ID
02	FoodRating	Integer	Range 1–5	Food rating
03	RiderRating	Integer	Range 1–5	Rider rating
04	Comment	String	NULL Allowed	Customer feedback
05	CreatedAt	DateTime	NOT NULL	Review date

2.2.21 Notification

Sr.	Name	Data Type	Constraint	Description
01	NotificationID	Integer	Primary Key, NOT NULL	Unique notification ID
02	Title	String	NOT NULL	Notification title
03	Message	String	NOT NULL	Notification content
04	Type	String	NOT NULL	Notification category
05	isRead	Boolean	DEFAULT FALSE	Read status
06	SentAt	DateTime	NOT NULL	Notification time

2.3. BUSINESS RULES

- A person can have only one role in the Foodsy system such as customer, admin, or rider.
- A customer must register an account before placing an order.
- A customer can have multiple delivery addresses.

- A customer can add multiple food items to the cart before checkout.
- Each menu item must belong to one category.
- A menu item can have multiple variants such as small, medium, or large.
- An order must contain at least one order item.
- A customer can place multiple orders, but each order belongs to only one customer.
- A promo code can only be applied if it is active and within the validity period.
- A promo code cannot be used after reaching its maximum usage limit.
- A bill must be generated for every confirmed order.
- A payment must be linked to only one bill.
- An order can only be assigned to one rider at a time.
- A rider can handle multiple deliveries, but only according to availability status.
- Delivery tracking information must store the current rider location with timestamp.
- A customer can only review an order after successful delivery.
- Ratings for food and riders must be between 1 and 5.
- Notifications can be sent to customers, admins, or riders for system updates.
- Cancelled orders must store a cancellation reason.
- Delivery status must be updated when the order is picked up or delivered.

2.4. ENTITY RELATIONSHIP DIAGRAM

The Entity Relationship Diagram (ERD) shown in Figure 1 illustrates the complete database structure of the Foodsy system. It shows all entities, their attributes, and the relationships between them. The diagram covers the connections between customers, orders, deliveries, payments, and other core modules of the system.



APPENDIX: AI USAGE AND PROMPTS

This section documents the use of AI tools during the preparation of this report. All AI-generated content was reviewed, verified, and adapted by the authors to ensure accuracy and alignment with the project requirements.

Sr.	AI Tool Used	Prompt / Purpose	Section Applied
01	Claude AI (Anthropic)	Generate the complete IEEE-format LaTeX report from the database design document including all data dictionaries, business rules, and entity descriptions.	All Chapters
02	Claude AI (Anthropic)	Format the report according to the document's heading and paragraph style guidelines (font, size, alignment, spacing).	Full Document
03	Claude AI (Anthropic)	Add placeholder and instructions for inserting the ERD image in the LaTeX report.	Section 2.4

Note: Claude AI (claude.ai) was used to assist with documentation formatting and LaTeX code generation. All technical content, entity definitions, business rules, and database design decisions were produced by the project team.